

A Two-Way Table Before Inference

AP Statistics · Unit 3 · Topic 3.14 · B: 2027-01-11 · E: 2027-01-11 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU VAR-8** — The chi-square distribution may be used to model variation.
- **LO VAR-8.H** — Calculate expected counts for two-way tables of categorical data. [Skill 3.A]
- **EK VAR-8.H.1** — The expected count in a particular cell of a two-way table of categorical data can be calculated using the formula: $\text{expected count} = (\text{row total})(\text{column total}) / \text{table total}$.

Essential question: Why is a chi-square condition judged from expected counts rather than the observed cells?

DOK-3 skill: 4.C · **FRQ pattern:** expected-count-boundary-and-remedy

DOK rationale: Part (c) coordinates the null-table calculation, a strict boundary condition, sampling evidence, and the meaning of possible remedies to adjudicate two claims and explain why the tempting observed-count rule is unreliable.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Display the board and ask for one sentence using the study description.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.
- Collect at the door. Score part (c) only using the three required elements.

Students do

- Write a one-sentence first take before any discussion.
- Work (a) and (b) from the rules box and the stem.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **expected-not-observed** (required) — Explains why expected counts rather than observed counts determine this check
- **boundary** (required) — Uses expected 5 and applies the printed greater-than-5 rule correctly
- **next-step** (required) — Suggests a larger random sample with rechecking, retaining the categories

E	Explains the count distinction, applies the printed boundary rule, and gives a next step preserving the categories.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] TODAY'S PROBLEM — annotated

Suppose a membership association has 5,000 new members and takes an SRS of 120. Each member is classified by orientation status and preferred support channel. The association may test whether the variables are associated.

Nico: “Every observed cell is at least 6, and the data came from an SRS, so the usual chi-square setup is ready.”

Priya: “The margins may produce a different boundary check; I would calculate that table before deciding.”

Support channel

Status	Chat	Email	Phone	Total
Done	42	34	24	100
Not done	8	6	6	20
Total	50	40	30	120

FIRST TAKE (5 min)

Would you decide from the recorded counts alone or also work out what the totals predict? Explain in one sentence.

Key: Ungraded. Everyday language is enough before the discussion; formal vocabulary belongs in the finish.

Rules callout “EXPECTED COUNTS UNDER NO ASSOCIATION (keep on the page)” is printed on the student sheet.

DOK 1 (a) Which row total, column total, and table total are needed for Not done/Phone?

Key: The row total is 20, the Phone column total is 30, and the table total is 120.

DOK 2 (b) Calculate the expected counts for Done/Chat and Not done/Phone. The other four expected counts are all greater than 5.

Key: Done/Chat: $(100)(50)/120 = 41.6667$. Not done/Phone: $(20)(30)/120 = 5$.

DOK 3 (c) In 3–4 sentences, decide whose claim follows the printed rule. Compare observed 6 with the expected Phone count, explain why the kind of count matters, and suggest one next step that keeps the same support-channel categories.

Key: Priya is justified: the rule checks expected counts, and the Not-done/Phone expected count is 5, which is not greater than 5. The observed count of 6 does not satisfy a rule about what the no-association model predicts. Under this sheet’s stated rule, do not proceed with the usual approximation yet. Collect a larger random sample and recheck the expected counts while retaining the same categories.

EXIT

One thing the rules box changed about my first take: _____